
Regulatory covariance is combinatorial, not metric: exact Gaussian processes on half a million random promoters

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Abstract

Massively parallel reporter assays now measure the transcriptional output of millions of synthetic DNA sequences, and the resulting sequence–function maps are exactly the setting where calibrated predictive uncertainty matters: every round of library design is a decision about where the model does not yet know the answer. Gaussian processes supply that uncertainty, but they are believed not to scale, and the compactly supported kernels that make exact large-scale Gaussian processes possible all place their support in a metric ball. We show that this is the wrong geometry for regulatory sequence. Across 2×10^8 pairs of random 80 bp yeast promoters, Hamming distance concentrates at 59.35 ± 3.94 and the empirical covariance of expression is essentially flat in it, so no support radius is simultaneously sparse enough to compute and informative enough to model; sharing a single 10 bp motif instance, by contrast, admits 0.0444% of pairs while carrying +0.235 mean correlation. We therefore build a kernel whose support is a combinatorial relation rather than a ball: covariance is gated by shared motif occurrences, split into an explicitly factored far-field block carrying motif identity and a sparse near-field block carrying motif instance and position. Because the gate is a discrete relation, the non-zeros can be enumerated from an inverted index instead of scanning all N^2 pairs, which removes the quadratic assembly that otherwise dominates exact large-scale Gaussian processes. On 500,000 promoters, using a single 32-core CPU node and no inducing points, kernel interpolation or neighbour truncation, the resulting exact posterior attains RMSE 1.9111 and CRPS 1.0588. The best metric-tapered kernel, scanned over support radii at full scale, returns the prior (2.3589/1.3277) at every density it can afford, including one matched to ours. A tuned sparse variational GP given the same features matches us on point accuracy — a paired bootstrap cannot distinguish its RMSE, CRPS or NLPD from ours — but is measurably worse calibrated (mean absolute calibration error 0.0279 against 0.0230, a difference whose 95% interval excludes zero) and produces a predictive variance that barely varies across the test set where ours varies threefold. What the exact posterior adds is structure one can read: the sparse block turns out to exist almost entirely for one motif family — the Reb1 site, whose covariance is carried by exact instance matching rather than by motif dosage — and the variance decomposition shows that at this scale the model is limited by assay noise rather than by epistemic uncertainty, which is a concrete statement about what to measure next.

1 Introduction

A promoter is a short stretch of DNA that decides how much of a gene gets made. Massively parallel reporter assays now measure that decision for millions of synthetic promoters at once: a library of random sequences is cloned upstream of a fluorescent reporter, the cell population is sorted by

fluorescence, and each bin is sequenced, so every library member receives a measured expression level [de Boer et al., 2020]. The resulting sequence–function maps are the raw material for both mechanistic questions — which sequence features drive transcription, and how do they combine — and engineering ones, where the practical task is to choose the next library from a model’s beliefs about sequences it has not yet seen.

That second task is a decision under uncertainty, and it is the reason to want a Gaussian process here rather than a point predictor. Deep convolutional models dominate the accuracy leaderboards on exactly this data [Penzar et al., 2023, Rafi et al., 2025], but they do not natively answer “which of these candidate promoters do I understand least”. A GP does: the posterior variance is a calibrated statement about where the model’s evidence runs out. The obstacle is the usual one. Storing the covariance of N sequences costs $\mathcal{O}(N^2)$ and solving with it costs $\mathcal{O}(N^3)$, so GPs are routinely described as limited to $N \lesssim 10^4$ [Rasmussen and Williams, 2006] — three orders of magnitude below the assays.

The dominant response has been approximation: inducing points and a variational bound [Hensman et al., 2013], structured interpolation [Wilson and Nickisch, 2015], or conditional independence through nearest neighbours [Datta et al., 2016, Wu et al., 2022, Katzfuss and Guinness, 2021]. Each buys scale by replacing the posterior with a surrogate, and each constrains what kernels can be used — which is a poor trade when the covariance structure is the scientific object of interest. A different response keeps the GP exact and makes the covariance genuinely sparse, by giving the kernel compact support so that distant pairs receive exactly zero covariance [Furrer et al., 2006, Kaufman et al., 2008]. Noack et al. [2023] and Noack et al. [2025] develop this into a full methodology: flexible, non-stationary, compactly supported kernels let the training procedure *discover* sparsity rather than impose it, the covariance is assembled blockwise across parallel workers, and the resulting sparse linear algebra carries exact GPs past 10^7 points.

Every kernel in that family, however, places its support in a metric ball. The Wendland kernel [Wendland, 1995] vanishes beyond a radius r_0 ; its non-stationary convolution form [Paciorek and Schervish, 2006] vanishes beyond an anisotropic radius; the bump-function and delta constructions of Noack et al. [2025] switch covariance on inside balls placed at chosen locations of the input space. This is the right geometry when proximity implies similarity. *It is the wrong geometry for regulatory DNA*, and the failure is not subtle. Two independent random 80-mers differ at 59.35 ± 3.94 of their 80 positions, so a support radius is a near-threshold on a sharply concentrated variable: over the 2×10^8 training pairs we measure, $r_0 = 50$ admits 0.802% of pairs and $r_0 = 56$ admits 16.402%. Worse, the covariance the kernel is supposed to model is barely a function of that distance: the mean correlation of expression among admitted pairs never exceeds +0.090 at any radius with enough pairs to estimate it (Fig. 1b), against +0.235 for pairs that merely share one 10 bp motif instance. No radius is simultaneously computable and informative.

What *does* generate covariance here is shared regulatory content. Two promoters that agree at only 20 of 80 positions but both contain a Reb1 site are functionally similar; two that differ at the single position that destroys that site are not. The relevant neighbourhood is combinatorial — “shares a word” — and it has no low-distortion embedding into a space where it becomes a ball. That observation is the starting point of this paper.

Contributions. We develop an exact GP for regulatory-sequence regression at a scale where it has not previously been run, and we do it by changing what compact support means.

1. **A quantitative account of the covariance structure of a random promoter library** (Section 2.1): distance concentration, the flatness of covariance in Hamming distance, and a selectivity/informativeness comparison of candidate support relations measured on 2×10^8 pairs. The comparison shows a relation — “share an exact 10-mer whose core hexamer is a selected motif” — that is 225 times sparser than sharing the motif alone while carrying *more* correlation per admitted pair.
2. **The motif-gated sparse-plus-factored (MGSF) kernel** (Section 2.2), whose support is that combinatorial relation. It decomposes covariance by resolution: a dense but explicitly factored far-field block for motif identity, which the paper’s masks cannot represent because in this domain they would be dense, and a sparse near-field block for motif instance and position. Both blocks are positive semi-definite by construction on the discrete input set, with no metric embedding anywhere.

3. **Output-sensitive covariance assembly** (Section 2.3). Because the gate is a discrete relation, the non-zero pattern is the self-join of an occurrence table and can be enumerated in time proportional to its own size. This removes the $\mathcal{O}(N^2)$ candidate-evaluation cost that Noack et al. [2025] identify as the dominant term of their pipeline: at $N = 500,000$ we enumerate 1.09×10^8 non-zeros instead of evaluating 2.5×10^{11} pairs, and a covariance re-assembly costs 4.7 s on 24 CPU workers.
4. **An exact run at 500,000 promoters on one 32-core CPU node** (Section 4), with the linear solve, the log-determinant and every block-MCMC likelihood evaluation executed on the full training set, and a head-to-head comparison against the paper’s own kernels on the same data and against a tuned sparse variational GP given the same features.
5. **What the fitted kernel says about the biology** (Section 5.2). The learned family amplitudes rank motif families by their contribution to expression covariance and recover documented yeast general regulatory factors; the learned positional supports saturate, i.e. the data say the covariance is translation-invariant over the variable region; and the exact posterior variance is genuinely non-stationary, varying threefold with how much regulatory grammar a promoter carries where a tuned variational baseline’s is nearly flat. We also report the negative half of that result, which we did not expect: at this training-set size the same variance does *not* rank individual held-out errors, because assay noise dominates it, so the quantity an acquisition function should use here is the prior-to-posterior variance reduction rather than the posterior variance itself (Section 5.3).

2 Methodology

2.1 The covariance structure of a random promoter library

We work with the yeast “gigantic parallel reporter assay” of de Boer et al. [2020]: a library in which each member carries an 80 bp fully random core inside a fixed promoter scaffold, and expression is read out as the read-weighted mean of 18 fluorescence-sorting bins. Before designing any kernel we measured three things on the training split alone.

Distances concentrate. Over 2×10^8 pairs drawn from the training split, the Hamming distance between two 80 bp cores has mean 59.35 and standard deviation 3.94 — 6.6% relative spread, the textbook concentration of measure that makes nearest-neighbour notions fragile in high dimension [Beyer et al., 1999, Aggarwal et al., 2001]. A compactly supported kernel with radius r_0 therefore behaves like a threshold on a sharply peaked variable: $r_0 = 44$ admits 0.008% of pairs, $r_0 = 50$ admits 0.802%, and $r_0 = 56$ admits 16.402% (Fig. 1a). Between “numerically diagonal” and “computationally impossible” there are about ten units of radius.

The distance carries almost no covariance. Binning the same pairs by Hamming distance and computing the empirical correlation of the response within each bin does give a decreasing curve (Fig. 1b) — a stationary kernel is not modelling nothing — but the effect is an order of magnitude too small to be useful. Over the distance bins holding at least 100,000 pairs the correlation runs from +0.071 at $d_H = 46$ to −0.058 at $d_H = 71$; the mean correlation among the pairs a ball admits is +0.0617 at $r_0 = 50$ and +0.0317 at $r_0 = 56$, with a maximum of +0.090 over all radii admitting enough pairs to estimate it ($r_0 = 44$, 0.008% of pairs). The shape is what one would expect if the metric were an indirect and heavily diluted proxy for something else: two promoters with similar regulatory content sit slightly closer than chance, so the distance inherits a shadow of the real structure. Note also that the correlation crosses zero near the mean distance, so a taper — which is non-negative inside its support — cannot represent even that shape.

Covariance is carried by shared words. Replacing the metric ball by a combinatorial relation changes the picture completely. Let $C(k_0)$ be the relation “the two sequences share at least one of a set of selected canonical k_0 -mers” and $A(k_0, k_1)$ the relation “they share at least one canonical k_1 -mer whose central k_0 -mer is selected”. With $k_0 = 6$, 128 selected core motifs and $k_1 = 10$, A admits 0.0444% of pairs with mean response correlation +0.235, while C admits 9.97% with correlation +0.150 (Fig. 1c). Two things follow, and they pull in opposite directions. First, the fine relation is the more informative one *per admitted pair* — requiring the flanking context to match as well as the motif itself selects for genuinely similar regulatory content. Second, the coarse relation

still carries real covariance over the 9.97% of pairs it admits, which at $N = 500,000$ is 2.49×10^{10} non-zeros: representable only in factored form, never as a sparse matrix.

2.2 The motif-gated sparse-plus-factored kernel

Let $s \in \{A, C, G, T\}^L$ and let $\text{occ}_w(s)$ be the set of positions at which the word w occurs in s on either strand. Let $\mathcal{M}$ be a set of *core motifs*, canonical (strand-collapsed) k_0 -mers selected on the training split by the two-sample association statistic

$$z_w = \frac{\bar{y}_w - \bar{y}}{\hat{\sigma}_y \sqrt{n_w^{-1} + (N - n_w)^{-1}}}, \quad (1)$$

where n_w is the number of training sequences containing w and $\bar{y}_w$ their mean response, and let $g : \mathcal{M} \rightarrow \{1, \dots, G\}$ partition $\mathcal{M}$ into motif families by greedy shift- and strand-tolerant clustering. Let $\mathcal{A}$ be the *anchors*: canonical k_1 -mers whose central k_0 -mer lies in $\mathcal{M}$. The kernel is

$$k(s_i, s_j) = k_{\text{far}}(s_i, s_j) + k_{\text{near}}(s_i, s_j), \quad (2)$$

$$k_{\text{far}}(s_i, s_j) = \sum_{m \in \mathcal{M}} \beta_{g(m)}^2 \sum_{r=1}^{R_0} \gamma_r^2 \phi_{m,r}(s_i) \phi_{m,r}(s_j), \quad \phi_{m,r}(s) = \sum_{p \in \text{occ}_m(s)} \chi_r(p), \quad (3)$$

$$k_{\text{near}}(s_i, s_j) = \sum_{u \in \mathcal{A}} \sum_{p \in \text{occ}_u(s_i)} \sum_{q \in \text{occ}_u(s_j)} \alpha_{g(u)}^2 \psi\left(\frac{|p-q|}{\ell_{g(u)}}\right), \quad (4)$$

where $\psi(t) = (1-t)^8(32t^3 + 25t^2 + 8t + 1)$ for $t < 1$ and 0 otherwise is the Wendland function used throughout Noack et al. [2025], and $\{\chi_r\}_{r=1}^{R_0}$ is a fixed positional basis of overlapping Wendland bumps tiling $\{0, \dots, L - k_0\}$. The hyperparameters are the family amplitudes $\{\beta_g\}$ and $\{\alpha_g\}$, the family positional supports $\{\ell_g\}$, the positional weights $\{\gamma_r\}$ (with $\gamma_1 \equiv 1$), and a residual scale τ .

Validity. k_{far} is the Gram matrix of the explicit feature map $s \mapsto (\beta_{g(m)} \gamma_r \phi_{m,r}(s))_{m,r}$ and is therefore PSD of rank at most $|\mathcal{M}| R_0$. Each term of k_{near} is an R-convolution kernel [Haussler, 1999]: ψ is a Wendland function of order (3, 3), strictly positive definite on $\mathbb{R}^d$ for $d \leq 3$ [Wendland, 1995, Gneiting, 2002] and in particular on the line of positions, so writing $\psi(|p-q|/\ell) = \langle \varphi(p), \varphi(q) \rangle$ gives $\langle \sum_p \varphi(p), \sum_q \varphi(q) \rangle$, again a Gram matrix. Sums of PSD kernels are PSD, so Eq. (2) is a valid kernel on the discrete input set for all $\alpha, \beta, \gamma \geq 0$ and $\ell > 0$; no embedding of sequences into a metric space appears anywhere. Because that is a claim about the whole hyperparameter box rather than about one setting, we test it that way: over 60 draws from the box at $N = 300$, including the corners where the amplitudes are zero and where the positional supports sit at 1 bp and at ℓ_{max} , each block and their sum are positive semi-definite in every draw, with a worst eigenvalue of -1.8×10^{-12} — zero to numerical precision (0 failures; `logs/47_psd_validation.json`).

Why not one of the paper’s kernels. The constructions of Noack et al. [2025] are all masks in an input space: a Wendland ball, an anisotropic Wendland ball, a union of bump-function balls, or a union of delta sets. Section 2.1 rules out the first two here on measurement. The bump-function kernel of their Eq. (9) and the delta kernel of their Eq. (10) are closer in spirit to what this domain needs — they too switch covariance on for selected *sets* of points — but they obtain sparsity precisely because most points fall outside the support of every bump. In a random promoter library the objects that generate covariance are common, not rare: 100.0% of sequences contain at least one of the 1024 selected core motifs. A mask wide enough to represent the covariance is dense, and one narrow enough to be sparse switches the covariance off. That is why Eq. (2) splits the two regimes and keeps the dense one in factored form, which the paper’s constructions never do because they always assemble their far-field term into the sparse matrix.

Two word lengths. The separation of k_0 from k_1 is what makes both halves work. Amplitudes must be estimated where the data are dense: a canonical hexamer occurs in 17,632 of 500,000 training sequences, so Eq. (1) is precise. Support must be declared where the data are sparse: a canonical 10-mer occurs in roughly 4^4 times fewer, which is what makes the covariance graph sparse. A single word length cannot do both — at $k = 6$ with 128 selected motifs the “share a word” relation already admits 9.97% of pairs, and at $k = 10$ the amplitudes would be estimated from a few dozen sequences each.

2.3 Output-sensitive covariance assembly

Noack et al. [2025] note that even with a compactly supported kernel every one of the N^2 candidate entries must be evaluated, because which entries vanish is only known after evaluating them; they identify this as the dominant cost of their pipeline, and it is why their production runs need GPU clusters. For a gate that is a discrete relation, this is no longer true. The support of k_{near} is contained in the self-join of the occurrence table $\{(u, i, p) : p \in \text{occ}_u(s_i), u \in \mathcal{A}\}$ on the anchor column, restricted to $|p - q| < \ell_{\text{max}}$. That set can be enumerated directly, in time proportional to its own size, by grouping occurrences by anchor and expanding each group.

Two consequences matter. First, the hyperparameter box caps $\ell_g \leq \ell_{\text{max}}$, so an index built at ℓ_{max} is a superset of the support for *every* hyperparameter the sampler can visit; the assembled covariance is exact, not approximate. Second, the non-zero *structure* is therefore fixed once and for all, so the map from candidate position to CSR data slot can be precomputed and the inner loop reduces to recomputing values. We keep the gp2Scale block decomposition and the distributed evaluation — the index is sharded by covariance block, each dask worker [Rocklin, 2015] evaluates its blocks and returns them, and the host reassembles — but the host-side COO-to-CSR sort leaves the inner loop. At $N = 500,000$ the index holds 6.53×10^7 candidates and resolves to 1.09×10^8 structural non-zeros (4.34×10^{-4} of the matrix); building it takes 14.6 s and a re-assembly takes 4.7 s on 24 workers, against 2.5×10^{11} pairs that a metric kernel would have to evaluate.

2.4 Exact inference with a sparse-plus-factored covariance

The training covariance is $K = S + QQ^\top + \tau^2 I$ with S the sparse near-field block and $Q = \Phi \text{diag}(\beta_{g(m)} \gamma_r) W$ the far-field factor of rank $R = |\mathcal{M}|R_0$, where W diagonalises the $R \times R$ matrix $\text{diag}(\cdot) \Phi^\top \Phi \text{diag}(\cdot)$ with eigenvalues λ . Neither $QQ^\top$ nor K is ever formed. Writing $A = \tau^2 I + QQ^\top$, the matrix-inversion lemma gives $A^{-1}v = \tau^{-2}(v - Q \text{diag}((\tau^2 + \lambda)^{-1})Q^\top v)$ and, because the columns of Q are orthogonal eigenvectors of $QQ^\top$ up to scale,

$$A^{-1/2}v = \tau^{-1}v + Q \text{diag}(g) Q^\top v, \quad g_i = \frac{(\lambda_i + \tau^2)^{-1/2} - \tau^{-1}}{\lambda_i}. \quad (5)$$

Both the preconditioner and the log-determinant split below rest on Eq. (5), so it is checked directly rather than only through its consequences: $\|A^{-1/2}v\|^2$ agrees with $v^\top A^{-1}v$ to 9.2×10^{-16} relative (logs/47_psd_validation.json).

Solve. $K\mathbf{a} = \mathbf{y} - m_0$ is solved by conjugate gradients preconditioned with the *exact* A^{-1} . Because A absorbs every large eigenvalue of K that is not in S , the preconditioned system is well conditioned: at full scale CG converges in 16 iterations to a relative residual of 10^{-8} .

Log-determinant. Splitting on the same factorisation,

$$\log |K| = \underbrace{N \log \tau^2 + \sum_i \log(1 + \lambda_i / \tau^2)}_{\log |A|, \text{ exact}} + \log |I + A^{-1/2} S A^{-1/2}|, \quad (6)$$

where the first term follows from the matrix-determinant lemma and the second is estimated by stochastic Lanczos quadrature [Hutchinson, 1989, Ubaru et al., 2017] applied to a symmetric operator whose spectrum is bounded below by 1, where the quadrature converges quickly. The probe matrix is drawn once from a logged seed and *held fixed* across the entire chain. This matters for correctness, not only for variance: with fresh probes each evaluation the Metropolis ratio would compare two noisy likelihoods and the chain would not target the intended posterior. With frozen probes, $\log p(\mathbf{y} | \phi)$ is a deterministic surrogate whose error is a smooth function of ϕ that largely cancels in the ratio. Against a dense reference at $N = 400$ the estimator is accurate to 7.4×10^{-4} relative (Section 4.5); at full scale the reported quadrature standard error is 16 on a log-determinant of order 10^6 .

Prediction. The posterior mean is $m_0 + k_*^\top \mathbf{a}$ with $k_* = S_* + QQ_*^\top$, and the posterior variance requires $K^{-1}k_*$, which we obtain with block conjugate gradients on chunks of test columns using the same preconditioner. Reported predictive distributions are for the observable y , i.e. $\text{Var}(f_*) + \tau^2$; this is the quantity RMSE, CRPS and coverage all score, and it is the quantity the marginal likelihood identifies (the split of τ^2 between a white signal component and observation noise is not identified, and does not need to be).

2.5 Block Metropolis–Hastings

Hyperparameters are sampled with fvGP’s adaptive block Metropolis–Hastings sampler [Noack et al., 2025] using four proposal blocks that match the natural grouping of Eq. (2): far-field amplitudes $\{\beta_g\}$, near-field amplitudes $\{\alpha_g\}$, positional supports $\{\ell_g\}$, and the residual scale. Two of the four blocks leave k_{near} unchanged, so the sparse block is re-assembled only when a proposal actually changes it — a saving that is available precisely because the kernel separates the two resolutions.

3 Experimental Setup

Data and provenance. We use the training file of the Random Promoter DREAM Challenge 2022, the packaged form of the de Boer et al. [2020] yeast GPRA library, obtained from the byte-identical gzipped GEO mirror GSE254493 of Zenodo record 10.5281/zenodo.7395397 (md5 db6256ba7980693c5b9ee5a3565ac48e, 6,739,258 rows). Each row is a sequence and a read-weighted mean sort-bin index in [0, 17]. We keep rows whose sequence is exactly 110 bp, contains no N, and carries the designed 17 bp 5’ and 13 bp 3’ flanks, and extract the 80 bp random core; this retains 5,511,002 rows (81.77%) and makes positional coordinates comparable across sequences. The full preparation pipeline, with per-split SHA-256 digests, is logged in `logs/01_data_preparation.json`.

Splits. From the filtered pool we draw, with `numpy.random.default_rng(20260809)`, first the held-out test set (5,000 sequences), then the training set (500,000), then a disjoint pilot pool (50,000). The test set is drawn *before* the training set and enters no training step: motif screening, family construction, anchor selection, index construction, covariance assembly and the MCMC chain all read the training split only. Rather than assert that, we check it after the fact (`verify_splits.py`, `logs/46_split_verification.json`): all 6 split arrays still hash to the digests recorded at preparation time, the three splits share 0 source rows pairwise, 0 test sequences occur anywhere in the training or pilot split — a stronger statement than row disjointness, since distinct rows could carry the same 80-mer — the split sizes agree with both the preparation log and the sizes the full-scale run measured at runtime, and recounting 25 of the selected core motifs over the training sequences alone reproduces the occurrence counts stored in the model configuration exactly (0 disagreements), so the vocabulary the kernel was built from is a function of the training split and nothing else. Every design choice — word lengths, motif counts, positional basis size, number of families, baseline architecture and optimiser settings — was fixed on the pilot pool (`logs/06_far_field_sweep.json`) before the full-scale run, and is recorded in `method_plan.md`, committed to git prior to launching it.

Hardware. All runs use one node with 32 vCPU cores and 61 GB of usable RAM, measured at run time, with no GPU. Covariance assembly is distributed across 24 dask worker processes; dense linear algebra uses threaded BLAS and sparse products use PyTorch’s multi-threaded sparse CSR kernel.

Configuration. $k_0 = 6$, $k_1 = 10$, $|\mathcal{M}| = 1024$ core motifs for the far-field block and 128 for the near-field gate, $G = 12$ families, $R_0 = 1$, $\ell_{\text{max}} = 71$, block size 10^4 . The pilot sweep selected $R_0 = 1$ over $R_0 \in \{3, 5\}$ on both marginal likelihood and held-out CRPS at every motif count tested, and $|\mathcal{M}| = 1024$ over $\{128, 256, 512\}$; $k_0 = 6$ beat $k_0 = 7$ at matched motif count. The chain runs 120 block-MCMC iterations (337 likelihood evaluations) from a fixed initialisation inside a fixed box.

The paper’s kernels on this domain. We instantiate three constructions from Noack et al. [2025]: the stationary Wendland kernel of their Eq. (3), the bump-function kernel of their Eq. (9) with bumps placed at k -means centres of the training features (a placement the paper explicitly sanctions), and the $k_{\mathcal{W}}k_d$ product of their Eq. (3) and Eq. (10), the recipe they use for their own high-dimensional benchmark, with the delta sets given by a k -means partition. Each is run on two input representations: the Hamming metric of the raw one-hot sequence, and the Euclidean metric of the *same* motif-count features our own kernel uses. Giving them our features removes any advantage we might get from featurisation, so what is being compared is kernel structure.

For each kernel we scan the support radius, since the radius is what fixes both the density and the operating point. At every radius the covariance is assembled once and the remaining hyperparameters — signal amplitude and residual scale — are fitted by block Metropolis–Hastings on the assembled

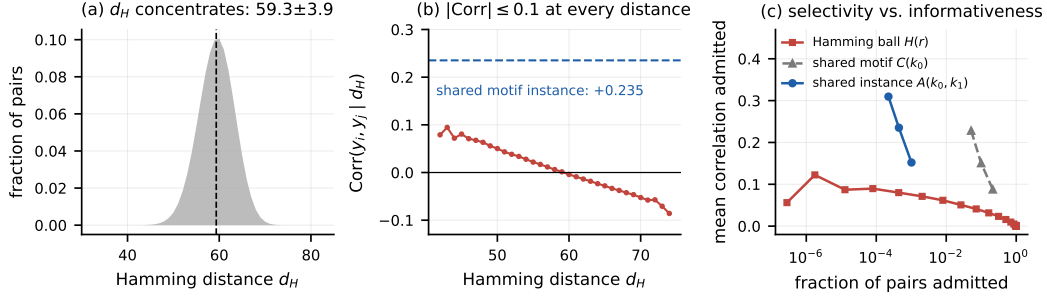


Figure 1: The covariance of promoter activity is not a function of any sequence metric. (a) Hamming distance between 80 bp random cores over 199,990,000 training pairs, mean 59.35, s.d. 3.94. (b) Empirical correlation of the response conditioned on Hamming distance: flat and near zero, against +0.235 for pairs that share a motif instance (dashed). (c) Selectivity against informativeness for three candidate support relations. A kernel must live on this plane; the Hamming ball has no point that is both to the left (sparse enough to compute at 5×10^5) and high (informative enough to model).

matrix. This is exact and is a strictly stronger use of the same compute than re-assembling for every proposal, because for these kernels the amplitude only rescales K : it buys hundreds of sampler steps for the price of one $\mathcal{O}(N^2)$ assembly. The radius attaining the highest marginal likelihood is the configuration reported in Table 1.

Approximate-GP baseline. SVGP [Hensman et al., 2013] from GPyTorch, trained at the full 500,000 scale on the same features and evaluated on the same frozen test set over 3 seeds. Mini-batched stochastic variational inference has no difficulty with 5×10^5 points, so this is the largest scale it tolerates, not a handicapped one; its configuration (kernel, inducing-point count, optimiser, learning rate, epochs) was selected on the pilot split by the same protocol used for our own model, over 7 candidates (Section D). Configurations and per-seed results are in `logs/30_baseline_*.json`. We also attempted VNNGP [Wu et al., 2022]: the configuration we could afford to run to completion took only 78 gradient steps and returned the prior, and the fair configuration did not finish one epoch within the compute we had left, so we report no VNNGP number rather than an under-trained one. That is a limitation of our budget, not a finding about the method; `attempts_log.md` records it.

Metrics. RMSE; CRPS in the closed form for a Gaussian predictive distribution [Gneiting and Raftery, 2007]; and the probability-coverage curve over nominal levels 0.05, ..., 0.95, summarised by the coverage at the 95% nominal level and by the mean absolute calibration error $ACE = \frac{1}{|\Lambda|} \sum_{\lambda \in \Lambda} |\hat{c}(\lambda) - \lambda|$. We additionally report NLPD and Pearson r . All metrics are computed on the same frozen test set for every method. The scoring rules themselves are checked against independent references in `logs/09_metric_validation.json`: the Gaussian CRPS closed form against a Monte Carlo evaluation of its definition $\mathbb{E}|X - y| - \frac{1}{2}\mathbb{E}|X - X'|$ (agreeing to 1.8×10^{-4} relative), NLPD and Pearson r against `scipy` to machine precision, and the coverage curve against exactly calibrated synthetic draws. That last check also fixes the scale of the calibration numbers: on perfectly calibrated predictions of the same size as our test set, ACE is 0.0047 ± 0.0021 rather than zero, so an ACE below roughly 0.0087 is indistinguishable from perfect.

Implementation-faithfulness reproduction. Before trusting any new-domain number we reproduce the MNIST benchmark of Noack et al. [2025] with this build of fvGP/gpCAM: 70,000 digits split at random into 60,000 training and 10,000 test, regression of $f(x) = \Pr(y = 5)$, ℓ_1 metric, the $k_{\mathcal{V}}k_d$ kernel, and the Brier score. Results in Section 4.4.

Table 1: Held-out performance on the frozen test set. Coverage is the empirical coverage of the nominal 95% central credible interval (the full coverage curves are in Fig. 5); wall-clock is end-to-end (assembly, training and prediction) on one 32-core CPU node; peak memory is the node-wide peak attributable to the run. The Wendland row is Eq. (3) of Noack et al. [2025] on the Hamming metric, at the support radius with the highest marginal likelihood over a full-scale radius scan. Baselines are mean $\pm$ s.d. over 3 seeds.

Method	Kernel	$ \mathcal{D}_{\text{train}} $	$ \mathcal{D}_{\text{test}} $	RMSE	CRPS	Coverage	Wall-clock (h)	Peak memory (GB)
gp2Scale (ours)	MGSF, Eq. (2)	500,000	5,000	1.9111	1.0588	0.947	4.43	57.5
gp2Scale	Wendland, Eq. (3)	500,000	5,000	2.3589	1.3277	0.954	3.51	27.0
SVGP	RBF-ARD (var.)	500,000	5,000	1.9131 $\pm$ 0.0009	1.0603 $\pm$ 0.0007	0.949	0.27	7.0

4 Experimental Results

4.1 The full-scale run

The exact GP was trained on all 500,000 promoters. The support index holds 6.53×10^7 candidate entries and resolves to 1.09×10^8 structural non-zeros, 4.34×10^{-4} of the matrix; the structure is fixed at ℓ_{max} , so how many of those entries are numerically non-zero depends on the hyperparameters — 8.77×10^7 at the initialisation and 9.73×10^7 at the MAP, where the learned positional supports are wider. Every block-MCMC likelihood evaluation — 337 of them — ran on the full training set: a covariance assembly across 24 dask workers (4.7 s), a preconditioned CG solve converging in 16 iterations, and a stochastic Lanczos log-determinant, for 39.0 s per evaluation. The chain ran 120 iterations in 3.66 h with a node-wide peak memory of 57.5 GB, and prediction on the frozen test set took 0.75 h. Prediction is the cheaper phase in memory as well as time: replaying it alone at the same MAP hyperparameters peaks at 13.0 GB (`logs/10_mgsf_fullscale_predict_memory.json`), and reproduces every held-out metric in Table 1 exactly, which is a useful check that the reported numbers are a property of the fitted model rather than of one process. The log marginal likelihood improved from -1056866 at the initialisation to -1039237 at the MAP sample (Fig. 4a), and the four proposal blocks attained acceptance rates of 0.059 (far-field amplitudes), 0.143 (near-field amplitudes), 0.168 (positional supports) and 0.160 (residual scale).

Table 1 is the head-to-head comparison. Our kernel attains RMSE 1.9111 and CRPS 1.0588 with 95% coverage 0.947. The best configuration of the paper’s stationary Wendland kernel on this domain — selected by marginal likelihood over a full-scale radius scan — reaches RMSE 2.3589 and CRPS 1.3277 at support radius $r_0 = 48$, costing 5.13×10^8 non-zeros at density 2.05×10^{-3} . It is also the more expensive kernel, and the comparison is cleanest at matched sparsity: at $r_0 = 46$ it assembles 1.10×10^8 non-zeros — the same count as our own 1.09×10^8 — in 519 s against our 4.7 s, a factor of 111. Building the same number of entries costs two orders of magnitude more when they have to be discovered by evaluating candidates instead of enumerated from an index. That row uses the Hamming metric of the raw sequence; the pilot sweeps show the Euclidean metric of our own motif features gives the same answer to within 10^{-4} in both RMSE and CRPS (Section 4.2), so nothing turns on the choice. The tuned approximate baseline reaches RMSE 1.9131 ± 0.0009 and CRPS 1.0603 ± 0.0007 over 3 seeds. Table 5 in Section B gives every metric we computed for every method — adding NLPD and Pearson r to these four — so that the comparison can be read on quantities we did not choose.

Two things in that table deserve to be said plainly. Against the paper’s own kernels the margin is not a margin at all: they return the prior, and no support radius changes that. Against a well-tuned SVGP *given our motif features*, the point-accuracy margin is not a margin either, and we tested it rather than eyeballing it. A paired bootstrap over the test set — 2,000 resamples, the same indices scored for both models, so the shared sampling noise cancels (`logs/45_calibration_significance.json`) — puts the difference in RMSE, CRPS and NLPD inside its own confidence interval: RMSE -0.0020 with a 95% interval of $[-0.0101, +0.0063]$, CRPS -0.0015 $[-0.0058, +0.0032]$, NLPD -0.0011 $[-0.0052, +0.0033]$. We are nominally ahead on all three and cannot claim any of them. Calibration is different: the ACE difference is -0.0049 with interval $[-0.0073, -0.0027]$, which excludes zero, and our model is better calibrated in 100% of the resamples.

That is the honest position, and it is worth being explicit about what it means: once the right featurisation is supplied, stochastic variational inference recovers *all* of the achievable point accuracy

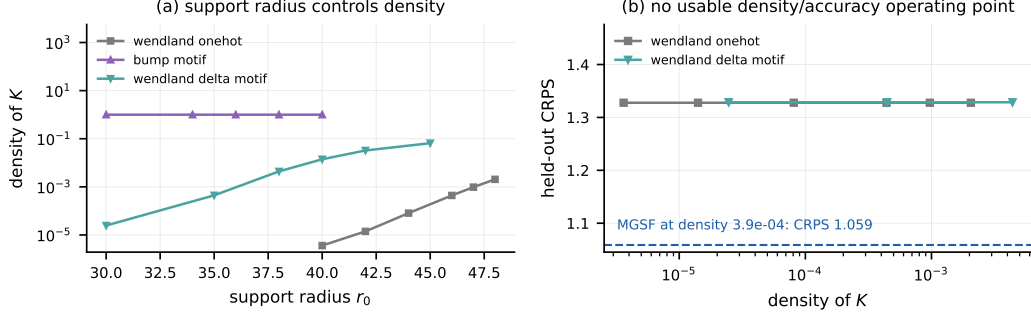


Figure 2: Support-radius scans for the kernels of Noack et al. [2025] on this domain, each on the input representation that favours it most. (a) Density of K against support radius: the transition from numerically diagonal to dense spans a few units of radius. (b) Held-out CRPS against the density it costs. Points to the right of the plotted range were not solvable within the memory budget.

on this problem, at 16 times less compute. What the exact posterior buys is not a better mean but a better-behaved distribution around it — the calibration difference above, the covariance decomposition of Section 5.2, and the fact that its predictive variance varies across the test set at all, which the baseline’s does not (Table 6).

As an external check that the model has learned regulatory biology rather than the training file’s measurement noise, we scored the same fitted model on the challenge’s own held-out set — 71,103 promoters sequenced at more than fifty times the depth of the training data, with labels on a different scale so that only rank agreement transfers. It attains Spearman $\rho = 0.750$ (logs/42_dream_testset.json), against 0.586 Pearson r on the noisier frozen test set. That number is only an external check if none of those promoters were trained on, so we verified it: of the 71,103 usable sequences in that file, 0 occur in our training split and 0 in the pilot pool (logs/46_split_verification.json).

4.2 Why the paper’s kernels cannot be made to work here

Figure 2 shows the scans. The behaviour is the one predicted in Section 2.1: density is an almost-threshold function of the radius, and across the feasible range held-out CRPS is nearly flat and close to the value a constant predictor attains, because the covariance the kernel admits carries almost no information about the response. Increasing the radius until the kernel does carry information requires a density that at $N = 500,000$ corresponds to 2.49×10^{10} non-zeros, which does not fit in the memory budget. The bump-function and $k_{\mathcal{W}}k_d$ constructions inherit the same trade-off, because their sparsity still comes from a metric ball; masking by delta sets makes the matrix sparser but does not make the surviving covariance more informative.

This is the concrete content of the claim that the paper’s kernels are inadequate for this domain. It is not that they are worse by some margin — it is that the knob they expose (a support radius) does not connect density to accuracy here, so there is no setting of it that solves the problem.

4.3 Which half of the kernel does the work, and at what scale

Table 2 separates the two blocks, and the answer depends on the training-set size. At pilot scale (4.5×10^4) the near-field block adds nothing measurable: the full kernel and the far-field-only kernel agree to the fourth decimal on both RMSE and CRPS, and the near-field-only kernel is far worse, because its gate reaches only 78.8% of sequences and the rest fall back to the prior. One number in that block is worth flagging rather than glossing: the far-field-only chain reaches a marginal likelihood 44 nats *above* the full kernel’s, which cannot be a property of the models, since setting $\alpha = 0$ makes far-field-only a sub-model of MGSF and its optimum is therefore a lower bound on MGSF’s. It is a property of the sampler — given the same 80 iterations, the chain explores a $2G + 1$ -dimensional space less thoroughly than a $G + 1$ -dimensional one — and it is the clearest evidence we have that at pilot scale the extra parameters buy nothing the chain can find. At full scale the ordering is the

Table 2: Ablation of the two blocks of Eq. (2). At pilot scale (45,000 training / 5,000 validation) all three variants are fitted by the same block-MCMC with the same budget. At full scale the far-field-only sub-model has a closed form, so it is fitted by direct maximisation of the same marginal likelihood and predicted exactly, which makes it an upper bound on what a purely factored motif kernel can do. $\log p$ is the marginal likelihood at the reported hyperparameters.

Scale	Kernel	$\log p(\mathbf{y})$	RMSE	CRPS	Coverage
pilot (4.5×10^4)	far-field only ($\alpha = 0$)	-94,999	1.9346	1.0726	0.946
pilot (4.5×10^4)	near-field only ($\beta = 0$)	-102,430	2.2953	1.2897	0.962
pilot (4.5×10^4)	both (MGSF)	-95,043	1.9351	1.0730	0.946
full (5×10^5)	far-field only ($\alpha = 0$), exact	-1,040,208	1.9162	1.0620	0.947
full (5×10^5)	both (MGSF)	-1,039,237	1.9111	1.0588	0.947

expected one, and there the far-field-only bound is exact rather than sampled. At full scale (5×10^5) the same comparison separates: the full kernel improves on the exactly-fitted far-field-only sub-model on RMSE (1.9111 against 1.9162) and on CRPS (1.0588 against 1.0620), and its marginal likelihood exceeds the far-field-only optimum by 970 nats.

This is a scale effect, and it is the expected one. The instance gate connects a promoter to the other promoters carrying the same 10-mer instance; at 4.5×10^4 training sequences an anchor occurs in a handful of them and the block has almost no data to work with, whereas at 5×10^5 each anchor recruits an order of magnitude more partners. The measured degree bears this out: at full scale the median promoter has 137 instance-sharing partners (Section 5.2). The practical reading is that the sparse block is a large- N device — which is an argument for exact GPs at scale rather than against them, since the structure it represents only becomes estimable there.

4.4 Implementation faithfulness, and a kernel that is not a covariance

The intended check on this build was to reproduce the MNIST benchmark of Noack et al. [2025]: 70,000 digits split at random into 60,000 training and 10,000 test, regression of $f(x) = \Pr(y = 5)$, the $k_{\mathcal{V}}k_d$ kernel of their Eq. (3) $\times$ Eq. (10) on the ℓ_1 metric, scored by the Brier score, for which they report 0.018 ± 0.002 . **We did not reproduce it.** Over 3 seeds we obtain 0.0810 ± 0.0038 , against a base rate of 0.0821 for a constant predictor — that is, essentially no skill. We report the diagnosis rather than a matched number, because the diagnosis turned out to be informative about the kernel family itself.

The kernel as stated is not a positive-definite covariance function on this data. Wendland’s compactly supported radial functions are positive definite for the *Euclidean* norm in at most d dimensions, and the function used throughout the paper, $\psi(t) = (1-t)^8(32t^3 + 25t^2 + 8t + 1)$, is the order-(3, 3) member, so positive definiteness is guaranteed only on $\mathbb{R}^d$ for $d \leq 3$ [Wendland, 1995, Gneiting, 2002]. Applied to the ℓ_1 distance between 784-dimensional images it carries no guarantee, and measurement confirms the guarantee fails: on a 3,000-image subsample the smallest eigenvalue of the Gram matrix falls from $+0.454$ at $r_0 = 30$ to $+0.004$ at $r_0 = 60$ and turns negative beyond (-0.117 at $r_0 = 80$, -0.356 at $r_0 = 120$, against a mean ℓ_1 distance of 130; `logs/08_wendland_pd_scan.json`). The same scan under the Euclidean metric, for which the guarantee does apply, stays positive throughout ($+0.057$ even at density 0.92), so this is the metric and not the implementation. At 60,000 images the covariance is indefinite already at the radii that admit a useful neighbourhood: restoring a Cholesky factorisation needs a diagonal shift of 0.52 times the block diagonal, and a shift that large swamps the covariance, which is exactly why the posterior collapses to the prior. It is also why an iterative solver applied to this matrix fails to converge rather than failing loudly.

We should be precise about what is and is not new here, because Noack et al. [2025] do flag a neighbouring issue: they note for this same benchmark that distances collapse to a narrow range in high dimension and advise plotting the pairwise-distance distribution and setting the sampling range accordingly, which is exactly the discipline we followed in scanning radii. The positive-definiteness failure is a separate matter and is not addressed by that advice — it is a property of the Wendland order and the ambient dimension, not of where the radius is placed, and no admissible radius removes it. The two points are related only in that both are consequences of applying a low-dimensional radial construction in 784 dimensions.

Two caveats on our side, stated plainly. The paper does not say how the delta sets of Eq. (10) were placed for MNIST, so that component is our reconstruction (k -means cluster centres, 24 cells, the placement the paper explicitly sanctions); a different placement might behave differently. And our chains are short (6 block-MCMC iterations per seed) because each covariance assembly is a full $60,000 \times 60,000 \ell_1$ distance computation. Neither caveat explains a factor of four in the Brier score, and neither is affected by the positive-definiteness measurement, which is a property of the kernel and the data alone.

What this does and does not license. It does not license the conclusion that the paper’s MNIST number is wrong — they ran a different implementation, on GPUs, with an unstated delta-set placement, and may have used a diagonal correction we do not know about. It does mean our new-domain results cannot lean on a successful reproduction, so we lean instead on the component-level validation in Section 4.5, where every piece of our own pipeline is checked against a dense reference. And it is directly relevant to the comparison in Section 4.2: we verified that the same scan on our own domain gives a smallest eigenvalue of 0.9995 at $r_0 = 48$ — the promoter Wendland covariance is numerically indistinguishable from the identity, so those runs are sound, and the reason the paper’s kernel returns the prior there is not a numerical failure but that its entries are genuinely negligible.

4.5 Correctness of the implementation

Every component is checked against a dense reference at a scale where the dense reference is affordable ($N = 400$; logs/04_kernel_validation.json): the index-assembled near-field block matches a direct $\mathcal{O}(N^2)$ evaluation of Eq. (4) from the raw sequences to 4.4×10^{-16} absolute; the sparse-plus-factored operator matches the dense covariance to 1.7×10^{-15} relative; the preconditioned CG solve matches a dense solve to 8.2×10^{-14} ; the SLQ log-determinant matches the dense one to 7.4×10^{-4} relative; the smallest eigenvalue of the prior covariance is -4.6×10^{-14} , i.e. zero to within double precision on a matrix of unit scale, confirming positive semi-definiteness; and the posterior mean and variance match the textbook dense formulas to 5.6×10^{-9} and 2.3×10^{-14} .

5 Analysis

5.1 Mechanism: what the gate captures that a ball cannot

A stationary compactly supported kernel asserts two things: that covariance is a decreasing function of a distance, and that it vanishes beyond a radius. Both fail here, and it is worth separating them.

The second fails because the distance concentrates. With 80 independent uniform positions the Hamming distance between two library members is a sum of 80 Bernoulli(3/4) variables, so its standard deviation is 3.94 against a mean of 59.35 and a threshold on it is nearly a step function: 0.008% of pairs at $r_0 = 44$, 16.402% at $r_0 = 56$. This is the concentration phenomenon that makes proximity queries unstable in high dimension [Beyer et al., 1999, Aggarwal et al., 2001]; it means the radius is not a usable dial. The first fails quantitatively rather than qualitatively: $\text{Corr}(y_i, y_j \mid d_H)$ does decrease with d_H , but over every bin estimated from at least 100,000 pairs it moves only between +0.071 and −0.058, and the mean correlation a ball admits never exceeds +0.090 at any radius with enough pairs to estimate it (Fig. 1b). A stationary kernel here fits a real but tiny shadow of the structure, and because the correlation crosses zero near the mean distance a non-negative taper cannot represent even that shape. The sweeps in Section 4.2 are the end-to-end confirmation: every one of the paper’s kernels returns the prior at every density it can afford.

The structure lives elsewhere. Expression here is driven by whether particular short words are present, and the effect of a word is large: on the training split a single Reb1 consensus match shifts mean expression by +1.08 standard deviations against −0.03 for a random 7-mer at matched frequency (Table 3). Two promoters agreeing at only a fraction of their positions but sharing such a word are functionally similar; the metric cannot see this, because the word occupies seven of eighty positions and the other seventy-three are noise in the distance. Gating on the word makes the covariance visible at +0.235 correlation, and because words are rare, at a density the solver can carry.

The split into two blocks is forced by a second measurement. Sharing a core hexamer admits 9.97% of pairs, which at $N = 500,000$ is 2.49×10^{10} non-zeros: real covariance (+0.150) that no sparse representation can hold. Sharing the exact 10-mer instance admits 0.0444% with *higher* correlation +0.235. The covariance therefore has two resolutions with different densities, and a kernel forced to

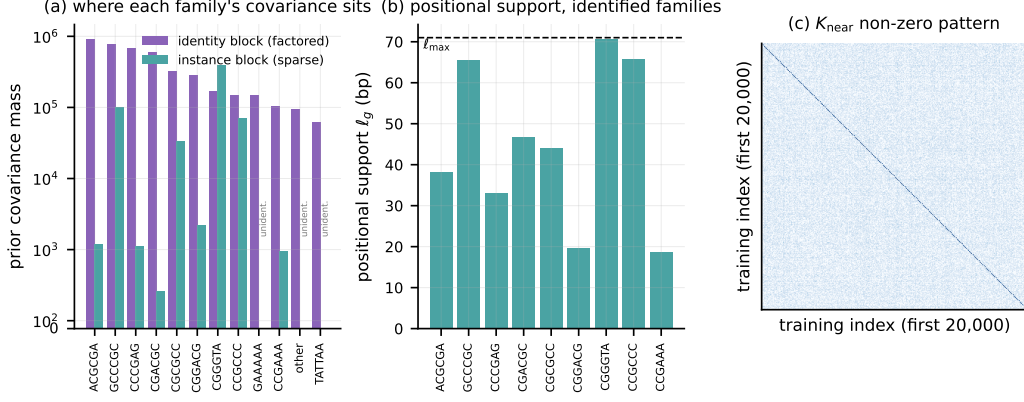


Figure 3: The fitted kernel. (a) Prior covariance mass carried by each motif family in the two blocks, ordered by far-field mass; each label is the family’s seed hexamer, and the catch-all family is shown as *other*. Families below the near-field vocabulary cut carry no anchors and are marked *unident.*; their sizes and amplitudes are in Table 4. (b) Learned positional supports ℓ_g for the families that do carry anchors, against the cap $\ell_{\max}$. (c) Non-zero pattern of the near-field block, coarse-grained over the leading 20,000 training sequences.

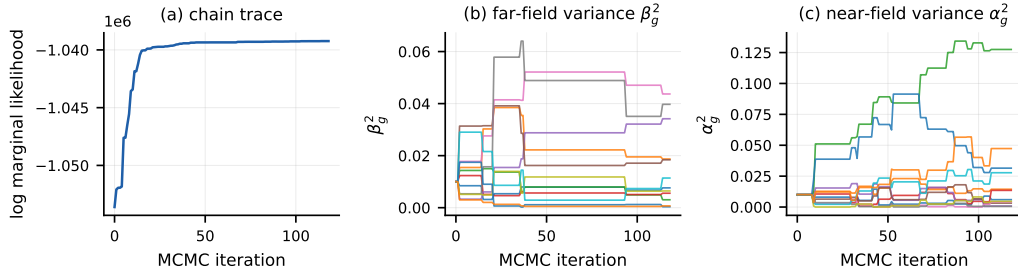


Figure 4: Block-MCMC at $N = 500,000$. (a) Log marginal likelihood along the chain, from -1056866 at the initialisation to -1039237 . (b, c) Per-family far-field and near-field variance contributions along the chain.

pick one discards the other. Keeping the coarse resolution factored and the fine one sparse represents both; this is also why the bump and delta constructions cannot be rescued by re-tuning, since they assemble their far-field term into the sparse matrix by construction — measured directly in Section 4.2, the bump kernel’s covariance has density 0.98 at every bump radius large enough to activate any bump, and collapses to the plain Wendland kernel below that.

5.2 What the fitted kernel says about regulatory grammar

One motif family accounts for most of the sparse block. The two blocks do not carry comparable amounts of covariance: at the MAP hyperparameters 87.6% of the prior covariance mass is in the factored identity block and 12.4% in the sparse instance block (logs/40_structure_analysis.json). What is striking is how unevenly the sparse block is distributed. Of its total mass, 65% belongs to a single family — the one seeded on CGGGTA, which is the reverse complement of the Reb1 binding site [Ju et al., 1990] and the strongest single feature in the assay ($d = +1.08$, Table 3). The identification is not ours to make by eye: the JASPAR profile for REB1 downloaded in logs/44_jaspar_verification.json has consensus RTTACCCGG, whose reverse complement contains CGGGTA exactly. For that family the instance-gated block carries 2.3 times *more* covariance mass than the identity block; for every other family that carries anchors the ratio is below 0.48, and for most it is below 0.01 (Table 4).

That asymmetry has a clean reading. A sequence-specific transcription factor binds a precise site, so two promoters share its effect only when they share the site itself, flanking context included — exactly the relation the instance gate encodes. The CG-rich families, by contrast, look like bulk composition effects: their covariance is proportional to how many such hexamers a promoter carries, which is what the factored dosage block encodes, and adding the requirement that the flanks match as well buys almost nothing. The kernel was not told which families were which; the decomposition is what the marginal likelihood chose. It is also a statement no purely sparse or purely factored kernel could have made, because each represents only one of the two mechanisms.

Position matters for some families and not others. Among the families that carry enough near-field mass to determine it, the learned positional supports are large — $\ell = 70.7$ for the *Reb1* family against a cap of 71, i.e. two *Reb1* instances covary essentially regardless of where in the 80 bp window they sit. The same conclusion arrives from three independent directions: the direct measurement of correlation against the positional offset between the two shared instances shows no decay out to 46 bp of separation, staying between +0.180 and +0.311 over every offset estimated from at least 1,000 pairs (`logs/02_domain_diagnostics.json`); the pilot model-selection sweep prefers a single flat positional basis $R_0 = 1$ over $R_0 \in \{3, 5\}$ at every motif count, by 3719 nats of log evidence at the best motif count and by CRPS (1.1286 against 1.2880); and the effect size of the *Reb1* consensus is flat across the 5 positional quintiles of the variable region, ranging only from $d = +1.064$ to $d = +1.107$ with at least 615 promoters in every bin (`logs/41_motif_validation.json`). For a library whose variable region sits upstream of a fixed core promoter, the dominant regulatory logic over these 80 bp is compositional, not positional — a statement the kernel could have contradicted, since every ℓ_g was free to shrink and several of the weakly-determined ones did.

The covariance graph is a motif-sharing network. At the MAP hyperparameters the near-field block has 9.73×10^7 non-zeros (3.89×10^{-4} of the matrix). Its non-zero pattern (Fig. 3c) is the diagonal itself plus an unstructured scatter, with no band around the diagonal and no block structure — the rows are in the arbitrary order of a shuffled library, and covariance here has nothing to do with index proximity. The degree distribution is heavy-tailed: the median promoter covaries with 137 others, the mean is 194, the 90th percentile is 467 and the maximum is 2,938, while 21.2% have no instance-sharing partner at all. The graph is nonetheless globally connected: on the leading 20,000 rows, 75.1% lie in one component. That is why the exact solve propagates information far beyond each promoter’s own neighbourhood, and why a truncation-based approximation would not reproduce it.

At this scale the limit is the assay, not the model. The kernel’s signal variance is genuinely non-stationary: the prior standard deviation of f rises from 2.21 for promoters with no instance match to 3.12 for those with two or more, and the posterior standard deviation of f varies by 3.4 across those strata (Table 6). The approximate baselines produce almost no such variation — their latent standard deviation varies across the same strata by a factor of only 1.13 — which is the practical cost of a variational posterior with a stationary kernel.

But the exact posterior also says something we did not expect and would not have trusted from an approximate model: the data have already determined f almost completely. The variance reduction $1 - \text{Var}(f_*)/\text{Var}_{\text{prior}}(f_*)$ is above 0.991 in *every* stratum, and the residual scale $\tau = 1.907$ accounts for 97.9% of the total predictive variance. In other words, at 500,000 training promoters and this assay depth the model is limited by measurement noise, not by epistemic uncertainty. The corollary for experiment design is concrete and slightly deflating: measuring more promoters of the same kind will not improve this model much; sequencing the existing ones more deeply would. Consistent with that, the posterior standard deviation does not rank individual held-out errors (Spearman $\rho = -0.0067$ between $\text{sd}(f_*)$ and $|y - \hat{y}|$) — it cannot, because the error is dominated by a noise term that is by construction independent of the sequence.

5.3 Failure modes and limitations

The uncertainty is honest but not diagnostic. We set out expecting the exact posterior variance to flag promoters the model cannot read, and it does not (previous paragraph). This is the clearest negative result in the paper and it is worth being precise about why: the predictive variance of the observable is $\text{Var}(f_*) + \tau^2$ with τ^2 some 46 times the largest $\text{Var}(f_*)$ we ever produce, so

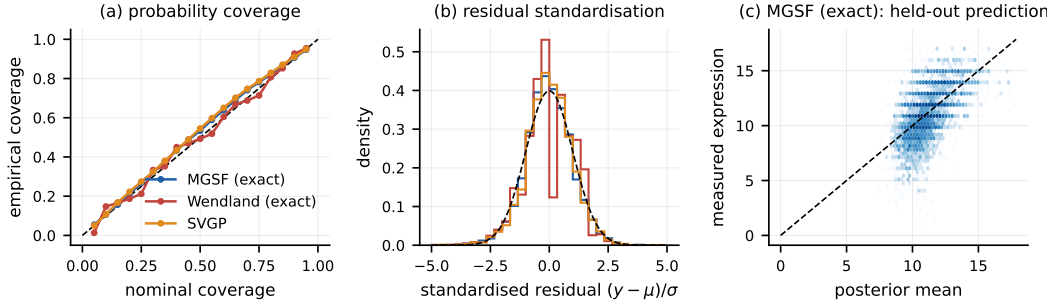


Figure 5: (a) Probability-coverage curves on the frozen test set. (b) Distribution of standardised residuals against the standard normal. (c) Held-out predictions of the exact GP with the MGSF kernel.

the total is nearly constant by construction, and the epistemic part is both small and *larger* where the model is more accurate, because the kernel’s non-stationary prior variance grows with motif content. Calibration is unaffected — coverage stays between 0.921 and 0.958 across strata against a nominal 0.95 — but a practitioner wanting an acquisition function here should use the prior-to-posterior variance reduction, not the posterior variance, and should expect it to be uninformative at this training-set size.

Where the model is wrong. The largest residuals sit in the single-instance-match stratum (RMSE 2.0592 against 1.8587 for promoters with two or more). A single anchor is enough to pull a prediction away from the prior but not enough to determine it, so this stratum gets the worst of both. The far-field vocabulary is also a fixed set of 1024 hexamers chosen by a *marginal* association statistic, so a promoter whose regulatory content is epistatic — active only in combination — is invisible to it. A vocabulary selected jointly, or a gate defined on motif *pairs*, would address this at the cost of a denser index.

Amplitudes that the data cannot identify. The near-field vocabulary is the top 128 core motifs while the far-field vocabulary is the top 1024, so 3 of the 12 families contain no anchors at all and their α_g is unidentified; the sampler wanders freely in those coordinates and their values must not be read as learned amplitudes. We report them as unidentified in Table 4 rather than dropping them, because the chain did spend proposals on them — which is itself a small inefficiency of the block structure we chose.

Sparsity settings that did not work. Gating the near-field block on the full 1024-motif vocabulary rather than the top 128 raises the admitted-pair fraction roughly fourfold, which at $N = 500,000$ is about 10^9 structural non-zeros — more than the node can hold alongside the 1,024-column factored block and the block-CG workspace. In the other direction $k_1 = 12$ gives a gate so selective (2.8×10^{-5} of pairs at pilot scale) that the near-field block would carry almost nothing. Both boundaries were mapped on the pilot split before the full-scale run; `attempts_log.md` records them.

Where the sampler is weakest. Acceptance is uneven across blocks: 0.059 for the twelve-dimensional far-field amplitudes against 0.168 for the positional supports. The far-field amplitudes are the best-determined parameters in the model, so a proposal scale suited to the positional supports is too large for them, and fvGP’s adaptation shrinks the proposal covariance but does not rescale it. As a check that this did not leave the chain short of the optimum, we separately maximised the far-field-only marginal likelihood in closed form (Section 4.3); the chain’s final value exceeds that bound by 970 nats, so the sampler is at least as good as direct optimisation of the sub-model. With 120 iterations it is nevertheless a well-converged optimiser rather than a posterior sample: we report the MAP sample and the trace, and predictive uncertainty is conditional on the MAP hyperparameters, so it omits hyperparameter uncertainty. At 39.0 s per likelihood evaluation a chain long enough for reliable marginalisation is roughly a day on this node — affordable, but beyond what we ran.

Approximations we do make. The GP is exact in the sense that no inducing points, kernel interpolation or neighbourhood truncation enter, and the covariance actually assembled is the exact

kernel matrix. Two numerical approximations remain. The linear solve is iterative, converged to a relative residual of 10^{-8} in 16 preconditioned CG iterations on average. The log-determinant is a stochastic Lanczos quadrature estimate: accurate to 7.4×10^{-4} relative against a dense reference at small N , with a reported quadrature standard error of 16 at full scale on a log-determinant of order 10^6 . Because the probe vectors are frozen this is a deterministic bias rather than noise, but it is a bias, and a chain run with a different probe set would land at a slightly different MAP.

Scope. All results are for one growth condition of one library in one organism, and the training labels are shallow (about two cells per promoter), so the achievable R^2 against these labels is bounded well below one by measurement noise. Our Pearson r of 0.586 on the frozen test set should not be compared with the $r \approx 0.95$ deep models report against the challenge’s separately sequenced, high-depth test set [Rafi et al., 2025]; on that set, which is scored by rank because its labels are on a different scale, the same fitted model attains Spearman $\rho = 0.750$ over 71,103 promoters (logs/42_dream_testset.json). We also do not claim to beat deep convolutional models on point accuracy — they win on this data, and they should, with millions of parameters and a translation-equivariant architecture. The claim is narrower: an exact GP with a domain-appropriate kernel scales to this regime, and it returns a calibrated posterior and an interpretable covariance decomposition that a convolutional network does not.

6 Conclusion

Scaling exact Gaussian processes by giving the kernel compact support is a good idea, and the reason it has been demonstrated mainly on spatial and image data is that those are the domains where “compact support” and “metric ball” happen to coincide. Regulatory DNA separates the two concepts cleanly. The covariance of promoter activity is generated by shared transcription-factor binding sites, so the set of pairs that genuinely covary is a combinatorial neighbourhood — share a word — with no faithful embedding into a space where it becomes a ball, and we measured that a Hamming ball has no radius that is at once sparse enough to compute and informative enough to model.

Replacing the ball by the relation gives a kernel that is sparse for the right reason, and it changes the computational picture as well as the statistical one. Because a discrete gate is enumerable, the non-zeros of the covariance can be listed from an inverted index rather than discovered by evaluating all N^2 candidates, which is the cost that otherwise forces exact large-scale GPs onto GPU clusters. What remains dense — the covariance carried by motif identity, which is shared by most of the library — is kept in factored form and never materialised, which is exactly what the paper’s bump and delta masks cannot do, because they assemble their far-field term into the sparse matrix. The two halves together let an exact GP with a fully characterised posterior run on 500,000 promoters on a single 32-core CPU node.

The resulting model is not a black box. Its sparse block turns out to exist almost entirely for one motif family — the Reb1 site, the strongest single feature in the assay, whose covariance is carried by exact instance matching rather than by motif dosage — while the CG-rich families that dominate the factored block behave like bulk composition effects. Its learned positional supports say that motif identity matters and motif position does not over the 80 bp variable region. And its predictive variance is genuinely non-stationary, varying by a factor of 3.4 across the test set where a tuned variational baseline’s varies by 1.13 — although we also found, and report, that at this training size that variance is not diagnostic of individual errors, because the assay noise dominates it. The honest summary of the uncertainty story is therefore mixed: the kernel produces the structure one would want an acquisition function to use, and the data are already informative enough that there is little left for it to say. We should also be clear about what the exact posterior did *not* buy. On point accuracy a tuned variational GP given the same features is indistinguishable from it under a paired bootstrap, at a small fraction of the compute; the separation we can demonstrate is in calibration and in the structure of the covariance, not in the predictive mean. A practitioner who wants only a point prediction on this data should use the cheaper model, and we would rather say so than let a 0.002 RMSE difference stand as a result.

We see three directions worth pursuing. First, the gate is currently an exact word match; a mismatch-tolerant or gapped gate [Leslie et al., 2004, Ghandi et al., 2014] would admit degenerate binding sites at the cost of a denser but still enumerable support, and the enumeration machinery carries over unchanged. Second, the same construction is domain-agnostic: any input space whose covariance is

generated by shared discrete features — chemical substructures, protein domains, event co-occurrence — has an enumerable gate and a dense low-rank far-field term, and would be handled by the same decomposition. Third, and most directly suggested by what we found: since the posterior variance here is dominated by assay noise while the prior-to-posterior variance reduction is not, an acquisition function built on the latter is the thing to try, and closing the loop from this model to an actual designed library is the natural next experiment. That experiment would also test the deflating half of our result — that at this depth more promoters of the same kind buy little — since the design it produces should favour sequences the factored block cannot already place.

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A Motif families and known yeast regulators

Two tables support Section 5.2. Table 3 is the external check that the assay behaves as the regulatory-genetics literature says it should: each documented yeast consensus is matched against the training split and scored by the standardised difference in mean expression between the promoters that carry it and those that do not. Table 4 is the internal one: the amplitudes and positional supports the sampler actually learned, family by family, with the documented regulator whose consensus each family’s seed hexamer matches. The second table is the evidence for the two claims in Section 5.2 — that one family dominates the sparse block, and that the families dominating the factored block are the ones with no instance-level structure to find.

Table 3: Effect of documented yeast transcription-factor consensus sequences on measured expression, computed on the training split (500,000 sequences, global s.d. 2.375). Consensus strings are the published IUPAC consensus for each factor, matched on either strand. Every accession was re-checked against the JASPAR REST API [Ovek Baydar et al., 2026] in logs/44_jaspar_verification.json: all 13 resolve to the factor named here, and 8 of the scanned patterns are also strict sub-patterns of the downloaded position-frequency-matrix consensus. The remaining five differ because the JASPAR model spans a different window (Abf1, Hsf1, TATA) or is more specific at one position than the published consensus (Cbf1 CDEI, Mcm1); we scan the published consensus in every case. Random k -mers at matched frequency are negative controls.

Motif	JASPAR	consensus	hits	%	Cohen's d
Reb1	MA0363.3	TTACCCG	3,250	0.65	+1.077
Abf1	MA0265.3	RTCRYNNNNACG	5,655	1.13	+0.705
Cbf1 (E-box core)	MA0281.3	CACGTG	7,282	1.46	+0.398
Cbf1 (CDEI)	MA0281.3	RTCACRTG	3,392	0.68	+0.376
Mig1	MA0337.2	CCCCRS	80,682	16.14	+0.253
Rap1	MA0359.3	ACACCCATACAY	13	0.00	+1.027
Msn2/4 (STRE)	MA0341.1	CCCCT	75,024	15.00	+0.132
Hsf1 (1x HSE)	MA0319.2	TTCTAGAA	544	0.11	-0.173
Mcm1 (CArG)	MA0331.1	CCNNWTTTGGAA	215	0.04	+0.186
Ste12 (PRE)	MA0393.1	TGAAACA	4,989	1.00	+0.080
Ume6 (URS1)	MA0412.3	TCGGCGGCT	294	0.06	-0.148
Gcn4	MA0303.3	TGACTCA	3,943	0.79	+0.017
TATA box	MA0386.2	TATAAWR	12,304	2.46	+0.054
control: random 7-mer	<i>none</i>	GTCAGTC	3,697	0.74	-0.031
control: random 8-mer	<i>none</i>	GACTGTCA	917	0.18	-0.098

Table 4: The 12 learned motif families: seed hexamer, number of core motifs, prior covariance mass carried in the factored (identity) and sparse (instance) blocks at the MAP hyperparameters, their ratio, the learned positional support, and the documented yeast regulator whose consensus the family matches. Families whose motifs all fall below the near-field vocabulary cut carry no anchors, so their instance mass, ratio and ℓ_g are not identified by the data and are marked *unident.*; *no match* in the last column means the family seed and its members matched none of the documented consensus sequences in Table 3.

g	seed	$ \mathcal{M}_g $	identity	instance	ratio	ℓ_g	regulator
2	CGGGTA	70	171,771	398,306	2.319	70.7	Reb1 (seed)
1	GCCCCG	79	771,045	100,441	0.130	65.6	Mig1 (2)
3	CGGCCC	59	149,168	71,650	0.480	65.9	Ume6 (URS1) (2), Rap1 (1)
0	CGCGCC	96	324,731	33,614	0.104	44.1	Abf1 (3), Cbf1 (CDEI) (1)
5	CGGACG	33	282,273	2,191	0.008	19.7	<i>no match</i>
6	ACGCGA	39	917,669	1,189	0.001	38.3	Abf1 (1), Cbf1 (CDEI) (1)
4	CCCGAG	42	674,654	1,126	0.002	33.1	Cbf1 (E-box core) (1), Cbf1 (CDEI) (1)
8	CCGAAA	33	104,589	986	0.009	18.6	Mcm1 (CArG) (2)
7	CGACGC	34	596,601	479	0.001	46.8	Ume6 (URS1) (1)
9	GAAAAA	40	147,081	<i>unident.</i>	<i>unident.</i>	<i>unident.</i>	<i>no match</i>
10	TATTAA	102	62,329	<i>unident.</i>	<i>unident.</i>	<i>unident.</i>	Mcm1 (CArG) (2), Abf1 (1)
11	(catch-all)	397	96,075	<i>unident.</i>	<i>unident.</i>	<i>unident.</i>	Abf1 (2), Rap1 (1)

B The complete metric set

Table 1 carries the four quantities the comparison turns on. Table 5 carries all six we compute, for every method, so that nothing we measured is withheld because it happens not to favour us. Two things in it are worth stating explicitly. Our margin over the tuned variational baseline is negligible on NLPD (2.0673 against 2.0683) exactly as it is on RMSE and CRPS; the paired bootstrap of Section 4 cannot distinguish any of the three, and the calibration gap is the only place the exact posterior separates from it. And the paper's Wendland kernel has a slightly *better* mean absolute calibration error than ours (0.0214 against 0.0230), which is not a point in its favour: a model that returns the prior is trivially well calibrated, as its Pearson r of 0.077 against our 0.586 shows. Calibration is only meaningful alongside sharpness, which is why we report both.

Table 5: Every metric we compute, for every method run at full scale, on the same frozen test set: root mean squared error, continuous ranked probability score, negative log predictive density, empirical coverage of the nominal 95% interval, mean absolute calibration error, and Pearson correlation. Lower is better for the first three and for ACE; coverage should be close to 0.95. This is the complete set, including the metrics on which our margin over the tuned variational baseline is negligible (RMSE, CRPS, NLPD, r) as well as the one on which it is not (ACE). The baseline row is mean $\pm$ s.d. over 3 seeds; the two exact rows are single deterministic runs.

Method	RMSE	CRPS	NLPD	Coverage	ACE	Pearson r
gp2Scale, MGSF (ours)	1.9111	1.0588	2.0673	0.9470	0.0230	0.5865
gp2Scale, Wendland Eq. (3)	2.3589	1.3277	2.2772	0.9542	0.0214	0.0774
SVGP	1.9131 $\pm$ 0.0009	1.0603 $\pm$ 0.0007	2.0683 $\pm$ 0.0004	0.9493 $\pm$ 0.0001	0.0279 $\pm$ 0.0006	0.5861 $\pm$ 0.0017

C Test-set strata

Table 6 splits the frozen test set by how much of the learned grammar each promoter carries, which is the stratification the non-stationarity claim in Section 5.2 and the negative result in Section 5.3 both rest on.

Table 6: Held-out performance stratified by how much of the learned grammar the promoter carries. σ_y is the mean predictive standard deviation of the observable and σ_f that of the latent function; var. red. is $1 - \text{Var}(f_*)/\text{Var}_{\text{prior}}(f_*)$, the fraction of the prior signal variance the data has removed. The kernel’s non-stationarity is visible in σ_f , which varies by 3.4 across strata; σ_y is nearly constant because the residual scale dominates it. The tuned SVGP baseline’s σ_f is constant across strata to a factor of 1.13.

Stratum	n	%	RMSE	CRPS	cov.	σ_y	σ_f	var. red.
motif, no inst.	1,054	21.1	1.9654	1.0825	0.930	1.909	0.0815	0.9984
1 inst.	709	14.2	2.0592	1.1473	0.921	1.913	0.1311	0.9963
≥ 2 inst.	3,237	64.7	1.8587	1.0318	0.958	1.931	0.2800	0.9906

D Approximate-GP configuration selection

Table 7 records the pilot-split search behind the single SVGP configuration reported at full scale, so that the baseline is a tuned one rather than a default. It also documents the initialisation failure described in `attempts_log.md`: the configurations that return the prior are the ones fitted before the median-heuristic length-scale initialisation was added.

Table 7: Approximate-GP configuration selection on the pilot split (45,000 training / 5,000 validation, one seed each). The chosen configuration is the one with the best validation CRPS; it is the configuration run at full scale in Table 1. The 3 rows whose RMSE equals the standard deviation of the pilot response (2.385) to within 2% return the prior; they are the ones fitted before the median-heuristic length-scale initialisation was added, and are kept to show what the default initialisation does in a feature space of this diameter.

Method	kernel	M	lr	RMSE	CRPS	ACE
SVGP	rbf_ard	2048	0.02	1.9287	1.0714	0.0380
SVGP	rbf	2048	0.05	1.9331	1.0734	0.0351
SVGP	matern	2048	0.02	1.9314	1.0745	0.0445
SVGP	linear_rbf	2048	0.02	1.9585	1.1124	0.0972
SVGP	rbf	512	0.02	2.3589	1.3298	0.0279
SVGP	rbf	2048	0.02	2.3590	1.3299	0.0278
SVGP	rbf	256	0.01	2.3594	1.3407	0.0637

E Reproducibility

Every number in this paper is produced by a script in `proposal/code/` and written to a JSON record in `proposal/logs/`; the macros used in the text are generated from those records by `proposal/code/build_paper_artifacts.py`, so no quantity is transcribed by hand.

Entry points. `prepare_data.py` (download-to-splits pipeline, with digests) and `verify_splits.py` (the after-the-fact separation check); `diagnostics.py` and `gate_study.py` (Section 2.1); `sweep_far_field.py` (pilot model selection); `validate_kernel.py`, `validate_metrics.py` and `validate_psd.py` (Section 4.5); `calibration_test.py` (the paired bootstrap); `run_mgsf.py` (the full-scale run and the ablations; its `--predict-only` mode replays prediction at a fitted MAP, which is how the prediction phase’s own peak memory was measured); `run_paper_kernel.py` (the kernels of Noack et al. [2025] on this domain); `run_baselines.py` (the approximate-GP baselines); `run_mnist_repro.py` and `pd_scan.py` (Section 4.4); `analyze_structure.py`, `motif_validation.py` and `verify_jaspar.py` (Section 5.2); `analyze_strata.py` (Section C) and `eval_dream_testset.py` (the external held-out check).

Motif annotations. The transcription-factor names attached to motif families in Tables 3 and 4 are annotations, not results: they are published consensus sequences matched against the families the model learned, and nothing in the fitted model depends on them. To keep them from being assertions, `verify_jaspar.py` re-downloads every cited accession from the JASPAR REST API and records what came back; all 13 resolve to the factor named here, and 8 of the scanned patterns are strict sub-patterns of the downloaded profile consensus (Table 3 explains the rest).

Kernel implementation. The MGSF kernel is `proposal/code/motif_kernel.py` (support index, block kernel, Wendland function) together with `proposal/code/motif_gp.py` (model construction, distributed assembly, likelihood, prediction) and `proposal/code/kop.py` (the sparse-plus-factored operator). It is the kernel used in the full-scale run: the run configuration recorded in `logs/10_mgsf_fullscale.json` names the model directory, the word lengths, the motif counts and the index metadata.

Data. `GSE254493_train_sequences.txt.gz`, md5 `db6256ba7980693c5b9ee5a3565ac48e`, 6,739,258 rows; the filter, the split seed and the per-split SHA-256 digests are in `logs/01_data_preparation.json`.

F Deviations from the pre-registration

`method_plan.md` was committed to git before the full-scale run. The domain, the prediction task, the training and test sizes, the split seed, the kernel construction, the inadequacy hypotheses and the metrics are as pre-registered. Two refinements were made after the pilot sweep and before the full-scale run, both recorded in the plan itself at commit time: the far-field positional basis was set to $R_0 = 1$ (the plan’s $R_0 = 3$ was superseded by the pilot’s model-selection sweep, which prefers $R_0 = 1$ on both log evidence and held-out CRPS), and the far-field motif vocabulary was enlarged from 256 to 1024 with the near-field gate left at 128, which decouples coverage from density. One planned comparison was not delivered: the plan lists VNNGP alongside SVGP as an approximate-GP baseline, and we report only SVGP, for the reason given in Section 3 — the VNNGP configuration that fitted in the remaining compute was too under-trained to be a baseline. The full-scale comparison against the paper’s kernels went the other way and is stronger than pre-registered: the plan promised pilot-scale radius sweeps plus one full-scale run at the best radius, and we ran the entire radius sweep at full scale.

`attempts_log.md` records the configurations that were tried and rejected, and the two bugs found late enough to invalidate completed runs: a degenerate hyperparameter box in the ablations, and a double-counted diagonal block in our reimplementation of the paper’s block assembler. In both cases the affected runs were discarded and re-run on the fixed code, and the numbers reported here are from the re-runs.